The Cost of Uneducated Children:

Externalities, Social Norms, and the Quantity-Quality Tradeoff

Abstract

My project develops a theoretical model of the quantity-quality tradeoff in fertility, incorporating negative externalities where a household's cost of education increases with the number of non-educating neighbors. I show that this can lead to multiple equilibria and that policies reducing the direct cost of children can counter-intuitively decrease total human capital by triggering a shift to a worse equilibrium or amplifying negative spillovers. These results, which hold for both discrete and continuous populations, call for caution in designing fertility-inducing policies.

1 Introduction

Economic theory tells us that the economy might gain from interventions in markets for goods with externalities. Recent literature (see e.g. Neidell and Waldfogel (2010)) shows that the market for children might actually have this property. Several papers (for example, Epple and Romano (1998)) study these effects in terms of the schooling market. At the same time, the effects of these externalities at the earlier stage of childbirth remain unstudied. This project is set to fill this gap.

I build a theoretical model of the quantity-quality tradeoff in a market with spillovers and an interaction between several households. My model mainly builds on Becker and Lewis (1973) discussed in class, but allows the cost of education to depend on the choices of neighbors. First, I consider a simple case with two counter-agents and discuss the comparative statics. Then, I take a continuum of agents and study spillover effects for different distributions of income across the households.

I find that the introduction of externalities can lead to the existence of multiple equilibria in a model with two neighbors. I also find that in some scenarios, a cheapening in the costs of a child can lead to a reduction in the total human capital of children born. Finally, I extend these results to the case with a continuous number of agents.

Children usually do not grow up on their own, but interact with others in kindergartens, schools, or when exercising their hobbies. It is quite likely that during these interactions they exchange thoughts, ideas, and also what economists would call quality (Hanushek et al., 2003; Neidell and Waldfogel, 2010). For example, an adolescent might be more likely to try drugs if someone from their friend circle is already doing so. This would be consistent with Lavy et al. (2012) who show that it is the peers at the worst end of the distribution that affect a schoolchild most.

Sometimes parents can influence and shape the friend group of their child, but this might include additional costs. For instance, Epple and Romano (1998) show in a theoretical paper that costly private schools attract peers of higher quality. Interestingly, people have been shown to be averse to migration (Clemens, 2011) which suggests that

parents are unlikely to avoid bad demographic trends. Hence, it appears reasonable that the cost of bringing up a child would depend on the average quality of children in the neighborhood or city. This might affect choices within the quantity-quality tradeoff.

Moreover, a society might develop norms that would affect the costs of bringing up children. For example, in a neighborhood where it is considered natural to have as many children as you can, it might also be very usual to co-organize the kids and ask for help with their upbringing. On the contrary, in a city where it is a norm to have few kids, asking neighbors to look after yours for an hour might result in very unpleasant questions.

Similarly, there could be norms about education. In countries where education is viewed as a norm, or even a right, basic schooling is literally costless for the parents, with the government taking over. In some countries with high norms it might even be costly to *not* give your child a proper chance to educate (e.g. in Germany this would be a crime). It might also be socially costly for you to give your child a worse education than 'the Joneses' from next door. At the same time, there are many countries where getting primary education is harder, since there is no social norm for that. For example, there are cases where even if you choose to send your child to school it might simply happen that there is no teacher present (Duflo et al., 2012).

This project might have some policy implications. The existence of market failures within the children market would express a need for taxes, laws, or other regulations (Randall, 1983). At the same time, this would be quite a sensitive topic, since, technically, it would mean that part of the decisions within the quantity-quality tradeoff would need to be made by the government. I point out that some fertility-inducing policies appear to ignore the redistributive effects and the changes in the composition of the population they bring about with themselves.

My project relates closely to literature on the quantity-quality tradeoff dating back to Becker and Lewis (1973) who provide a detailed theoretical setup, showing that, depending on the income elasticities of quantity and quality, an increase in income can lead to a reduction in fertility. Becker and Tomes (1976) later extend the model to show that it is the human capital production function combined with heterogeneous initial endowment that plays a crucial role in the negative relation of income and childbirth. Recent empirical work shows evidence for the mechanisms suggested. Li et al. (2008) find a heterogeneous negative effect of family size on educational attainment in rural and urban China by using twin birth as an instrument. Additionally, several papers like Hanushek (1992) suggest that the child order is also important for educational choices. My work extends this framework by examining a differential cost in heterogeneous environments with the quality of the peers of children affecting tradeoff decisions.

My paper is also connected to a growing literature on peer effects and spillovers. Here, the works of Hanushek et al. (2003) and Neidell and Waldfogel (2010) suggest strong evidence for externalities within classrooms. By using panel data and controlling

for several levels of fixed effects, the first paper finds that peer achievement has a positive effect on personal achievement growth, while the latter provides evidence that ignoring spillover effects would significantly understate the social returns to preschool. Lavy et al. (2012) extend the study to show that it is the lower tail of the ability distribution of peers that plays a crucial role, while they fail to find evidence that good peers would provide positive externalities. Finally, Nicoletti and Rabe (2019) turn specifically to sibling spillover effects and find that these account for a large proportion of the attainment gap between low- and higher-income pupils. All these papers study the schooling market and some suggest policy implications. My project goes a step back and considers decisions at the earlier stages of the quantity-quality tradeoff.

There is some literature concerning social norms and how they work. A very simple framework is presented in a work by Lindbeck (1997). There are two empirical papers by Namoro and Roushdy (2009) and Wahba (2006) that look at social norms relative to women empowerment and child labor accordingly. However, I struggle to find any theoretical papers modeling social norms regarding fertility. Moreover, I did not see papers that would model two-dimensional norms as for quantity and quality.

Finally, my study would contribute to more recent work related to government intervention in child markets. An important reference here is Wang and Zhang (2018) who examine the One-Child Policy in China. They find that a differential implementation of the policy across rural and urban areas leads to a change in the composition of the population and therefore actually reduces the average human capital by approximately 1–2%. This further stresses the need for caution about government intervention in such sensitive matters. Baudin (2012) points out that settings with an unknown number of surviving children differ substantially, which might suggest that policies in developing countries should also be different. My paper extends the study by showing that subsidizing low-income low-ability fertility could have negative effects on high-income households.

2 Theory

2.1 Households

Let us consider a modification of the very simple quantity-quality tradeoff model. Namely, there will be two consumers $i = 1,2$ who maximize their utility functions

$$\max_{n_i, e_i} n_i q(e_i), \;\; s.t. \; p_n^i n_i + p_q^i n_i e_i \leqslant w_i \tag{1}$$

where conventionally n is the number of children, e is education and q the quality of children, w is the wage, while p_j is the price of commodity j. Note that I leave out consumption as I do not care about it too much and since the log-linear utility function makes it constant anyway.

What will be new here compared to usual models, is that I will allow the price of quality to depend on the decisions of the neighbor. This might reflect both social norms,

and different types of externalities. It should be noted that in the theory developed by Lindbeck (1997) norms would usually enter the utility function, while I find it more convenient to introduce them into the cost function.

In accordance with classical models like Becker and Lewis (1973) I will set $p_n^i = \tau w_i$. I will make the model extremely simple and allow only for two possible values of e. A household can either choose to educate ($e = 1$) or to not educate ($e = 0$). The quality of each child will then be accordingly:

$$\begin{cases} q(0) = 1 \\ q(1) = 1 + h \end{cases}$$

Intuitively, say that each child gets a compulsory primary education, but the parents decide whether to send them to secondary school. The price of education will be

$$\begin{cases} p_q^i(0) = 0 \\ p_q^i(1) = \beta + \alpha(1 - e_{-i})n_{-i} \end{cases}$$

So primary education is costless, say, covered by the state, while secondary education induces a basic cost of β plus some additional costs if the neighbors decide to not educate. Technically, I introduce negative externalities of bad education. I could potentially add a symmetric positive externality, but I decided to keep the model simple.

There are several explanations for that externality in real life. First, bad quality children could provoke their neighbors to skip school and instead go hang out with them and do bad things, increasing the direct costs of education. Second, children that are not getting educated are available for labor, so they might set a norm for adolescent labor, increasing the alternative costs of those getting the education. Similarly, uneducated children have more available effort to spend in the marriage market, making the alternative costs for those people getting the education even higher. Finally, the more children are not getting educated, the more the norm shifts away from schooling. This might lead to less support from the government or the family, as well as more costly and less abundant private secondary schools or universities.

Intuitively, in some settings, it might make more sense to allow bad quality to affect the return to education (say $h(e_{-i},n_{-i})$) instead of the costs. Note, however, that this would not change the model too much. Each additional unit of education bringing γ less returns is equivalent to saying that one additional unit of quality became $\frac{1}{1-\gamma}$ more costly.

Note that in the costs function that I specify above, the negative externality α is amplified by the number of bad-quality children, which makes sense in terms of relative sizes of families. Technically, social norms shift away from education faster if there are more uneducated people. Similarly, the alternative costs of schooling become higher the

more people are getting experience in the labor market instead.

Note also that the presence of only the negative externality is justified by findings from Lavy et al. (2012). It would be easy to add symmetric positive externalities to the model, but this would not crucially affect the results, so I decide not to clutter the model too heavily with that.

2.2 Note on the Assumptions

One serious assumption here is that social norms affect costs continuously and differentiably. Note that this is rather not the case in real life. For example, in Russia there is a threshold policy, where all families with more than three children (the 'norm') have serious subsidies. Social sanctions could also be discontinuous. For example, there could be a stigma against childless families but not families with ε children. This assumption, however, simplifies the math significantly, so I decide to stick with it.

Note also that I do not restrict the norms to be strictly positive, i.e. when having as many children as possible is regarded socially desirable. There could always be social norms against having too many children. However, the point of my model is that these norms would be relaxed depending on the neighbors' behavior, while sanctions against not having enough children would increase.

2.3 Model in Game Form

I can write the household problems explicitly as

$$\max_{n_i \in \mathbb{R}_+,\ e_i \in \{0,1\}} n_i q(e_i),\ \ s.t.\ \tau w_i n_i + (\beta + \alpha(1 - e_{-i})n_{-i})n_i e_i \leqslant w_i$$

Clearly, the budget constraint binds, and clearly, for both agents, there are only two cases to consider. They can explicitly calculate the number of children they can get for each level of education and simply compare the derived utility.

Note that we can calculate the number of children as $n_i = \frac{1}{\tau}$ for households choosing to not educate, as $n_i = \frac{1}{\tau + \frac{\beta}{w_i}}$ if both households choose to educate and $n_i = \frac{1}{\tau + \frac{\beta + \alpha n_{-i}}{w_i}}$ if household i decides to educate while $-i$ decides not to do so.

	N	E
N	$\left(\frac{1}{\tau}, \frac{1}{\tau}\right)$	$\left(\frac{1}{\tau}, \frac{1+h}{\tau + \frac{\beta + \frac{\alpha}{\tau}}{w_2}}\right)$
E	$\left(\frac{1+h}{\tau + \frac{\beta + \frac{\alpha}{\tau}}{w_1}}, \frac{1}{\tau}\right)$	$\left(\frac{1+h}{\tau + \frac{\beta}{w_1}}, \frac{1+h}{\tau + \frac{\beta}{w_2}}\right)$

Table 1: The Model in Game Form

We can represent household choices in the form of a simultaneous game. Table 1 shows that game in the simple form. Here, N stands for the choice to not educate, i.e. send the children only to primary school, while E stands for the choice to educate, i.e. send the children to secondary school or university.

Figure 1 shows the utility, or total child human capital, of a household under the two choice options. The red line represents the total human capital when the household chooses not to educate their children, while the other two lines represent the total human capital when the household chooses to educate. The green line shows the utility when the households' neighbor chooses to educate, while the blue line shows the case when they choose not to.

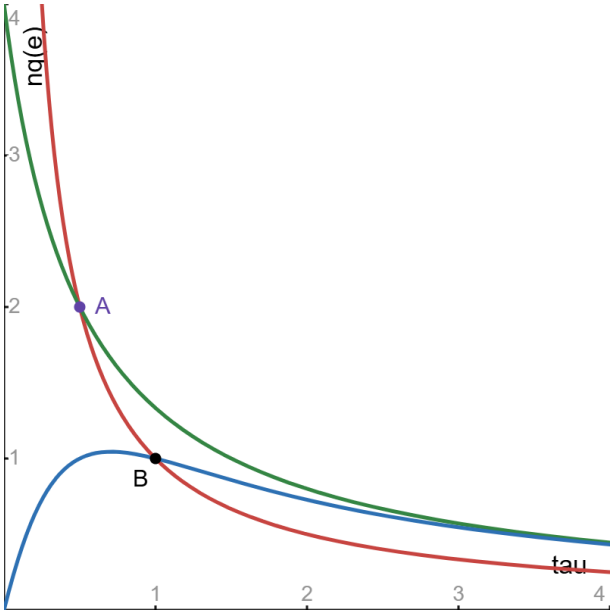


Figure 1: Consumer Utility under two Choice Options

A graphical representation of the negative externality is the downward shift of the blue line relative to the green line. Note that there are two important points labeled as A and B in the figure. Each point represents a transition from quantity to quality for the two corresponding cases. The following proposition addresses this formally.

Proposition 1. *The household follows a threshold strategy. That is, there exists some* $\bar{\tau}$ *such that for all* $\tau < \bar{\tau}$ *they do not educate, while for all* $\tau > \bar{\tau}$ *they choose to do so.*

This proposition, like most of the findings of this project, is quite straightforward and will not be followed by a formal proof or mathematical derivation.

Note that the threshold strategies of each household actually depend on each other. That is, relative wages are important for determining whether a households' comparative statics switching point would be at point A or point B from Figure 1. In what follows I describe the potential cases in detail.

2.4 Optimal Choices

Without loss of generality, assume $w_1 > w_2$. There are several cases to consider.

Proposition 2. *If* $h < \frac{\beta}{\tau w_1} < \frac{\beta}{\tau w_2}$ *holds, or if* $\frac{\beta}{\tau w_1} < h < \frac{\beta+\frac{\alpha}{\tau}}{\tau w_1}$ *holds together with* $h < \frac{\beta}{\tau w_2}$, *both households choose not to educate.*

In the first case, the relative benefit of education is too low for both households, while in the second case, household 1 would potentially want to educate, but the choice of household 2 makes that option too costly.

Proposition 3. *If* $h > \frac{\beta+\frac{\alpha}{\tau}}{\tau w_2} > \frac{\beta+\frac{\alpha}{\tau}}{\tau w_1}$ *holds, or* $h > \frac{\beta+\frac{\alpha}{\tau}}{\tau w_1}$ *holds together with* $\frac{\beta}{\tau w_2} < h < \frac{\beta+\frac{\alpha}{\tau}}{\tau w_2}$, *both households always choose to educate.*

Here, in the first case, the relative benefit is high enough anyway, while in the second case, the first household carries the economy towards the educated equilibrium.

Proposition 4. *If* $\frac{\beta+\frac{\alpha}{\tau}}{\tau w_1} < h < \frac{\beta}{\tau w_2}$, *the first household educates, but the second does not.*

Finally, the more interesting case

Proposition 5. *If* $\frac{\beta}{\tau w_1} < h < \frac{\beta+\frac{\alpha}{\tau}}{\tau w_1}$, *and* $\frac{\beta}{\tau w_2} < h < \frac{\beta+\frac{\alpha}{\tau}}{\tau w_2}$, *there are two equilibria, where either both households choose to educate or both do not.*

Note that the difference between Propositions 2 and 3 rather addresses the comparative statics of the cost τ, while the difference between 3 and 5 rather concerns the comparative statics of $\frac{w_1}{w_2}$. In Figure 2, I provide a case when Proposition 5 might apply.

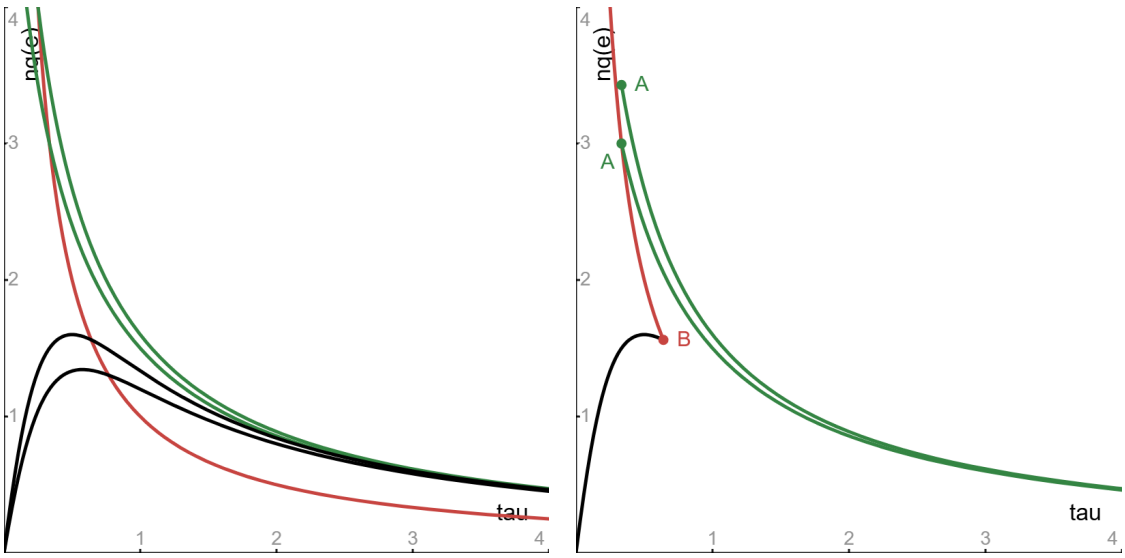


Figure 2: Two Consumers Potential Transition Points

The left figure shows the total human capital for each case for two households. Note that if τ is larger than at point B, both households necessarily educate. If τ is smaller than at point A, both choose not to educate. Technically, for such a situation to occur, we must have an interval of τ, where $\frac{\beta}{\tau w_2} < h < \frac{\beta+\frac{\alpha}{\tau}}{\tau w_1}$, which is possible if $\triangle w < \alpha h \frac{w_2^2}{\beta^2}$.

2.5 Government

One of the goals of this project is to show that government intervention might not always deal correctly with regulating such types of markets, and that sometimes it could even create harmful distortions.

Let the government consist of old selfish people that only care about the total human capital in an economy ($\sum n_i q_i$) to collect as many taxes as they can in the future. Suppose also that the government can somehow affect the costs of bringing up a child, namely, τ. For example, it could introduce free healthcare and free transportation, as well as other benefits not related to schooling. Perhaps, an introduction of parental leave might also be a possible policy here.

More importantly, the government could run a campaign trying to alter norms towards the number of children. This could literally be a one-child policy like in China, or it could be something more subtle, like introducing a social stigma against childlessness. This rather relates to the fields of signals or even marketing, so I will not go into detail with the mechanisms potentially to be in play.

For simplicity, let us define a subspace of available childcare policies as $\Theta = [\underline{\tau}, \overline{\tau}]$. Let also the exercise of this ability be costless. These two assumptions cancel out to some extent, as it might indeed be costless to start a pro-childbirth marketing campaign, but with the effects of the marketing campaign being rather limited and definitely bounded. At the same time, a significant change in τ might perhaps be so costly that no government would exercise it. I show that even in such cases, without policy costs and dead-weight losses, wrong interventions might be hurtful to the economy.

Suppose that the economy is in a situation similar to Figure 1 in between points A and B, and that the equilibrium behavior is that both agents decide to educate their children. Suppose also that the policy bounds are somewhere around point A. Figure 3 below suggests a graphical representation of this case, depicting the average human capital of the households' children, as well as the limits of the government.

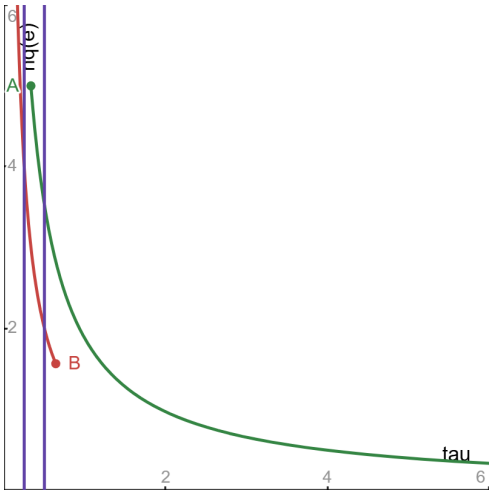


Figure 3: Average Human Capital and Intervention

Suppose that the initial position is exactly at point A and that the government tries to increase the number of children by intervening to slightly decrease τ. Note that the total human capital of the second household actually increases slightly. However, the human capital of the first household will decrease in a jump-like fashion. Hence, a decreasing intervention into τ will actually *decrease* the total human capital of the economy.

Now the government might be able to decrease the cost of children even further, but still at the limit there would be less total human capital. Suppose now the government would want to reverse the situation and start increasing τ. Note, however, that now the economy is in the 'bad' equilibrium, and it might require more effort to get back, e.g. get to point B first. Now note that the government might this time hit the upper bound and be forever stuck in the bad equilibrium. To sum up,

Proposition 6. *It is possible to shift the economy from good equilibria to bad equilibria by exercising policies not carefully enough. Moreover, it sometimes might not be possible to shift the equilibrium back with policies.*

Moreover, the government might not only decrease the total amount of human capital, it might in fact decrease the amount of children of the rich group by decreasing the cost of bearing children. This sounds counterintuitive, but is easily explained by the externalities.

Consider the case from 4 where the rich get educated, while the poor do not get educated. Note that, if the bounds are slack, the number of children in household 1 changes with τ as

$$n_1'(\tau) = \frac{-1 + \frac{\alpha}{\tau^2 w_1}}{\left(\tau + \frac{\beta + \frac{\alpha}{\tau}}{w_1}\right)^2}$$

Note that this derivative is positive if $\tau^2 < \frac{\alpha}{w_1}$. Hence,

Proposition 7. *If the economy is initially in the state from Proposition 4 and $\tau^2 < \frac{\alpha}{w_1}$, a decrease in the cost of children decreases the amount of children in the rich group.*

Intuitively, in some cases, the decrease of direct costs might be smaller than the indirect increase in the negative externalities from the poor family producing low-quality children.

Note that such policies heavily affect the demographic composition of the population. They might result in a severe change in the share of educated people in the economy. Dynamically, this effect might explode, since a larger share of poor families would even more increase the cost of education.

I am not claiming that the effects described above are indeed in place in our world. However, I would want to point out that a government should be aware of the reasonably plausible potential pitfalls a child-promoting policy might cause. I would call for a very careful consideration of any policy affecting the demographic composition of a country.

2.6 Multiple (Continuous) Consumers

In this Subsection I show that the results hold and might even be amplified in the case when there is a continuum of households as opposed to only two neighbors.

Suppose that there is a unit number of consumers and that we know the distribution of their income. I will consider two cases in particular. First, I turn to the uniform distribution. Then, I will examine the exponential distribution of wages that might be more realistic for our world.

Let the distribution of parent income have cdf $F(w)$ and pdf $f(w)$. Let also the price of education depend on the share of not educating *households* as opposed to uneducated children as it was above. Namely, let

$$p_e = \beta + \alpha x, \text{ where } x \text{ is the share of non-educating household}$$

Let each household maximize the total human capital as described above. In order to find some equilibria in this economy let us follow a few simple steps.

Proposition 8. *If some household with wealth $\underline{w}$ chooses not to educate, then all households with $w < \underline{w}$ also choose not to.*

Proof. Note that each household compares $\frac{1}{\tau}$ to $\frac{1+h}{\tau+\frac{\beta+\alpha x}{w}}$. Note that if $\frac{1}{\tau} > \frac{1+h}{\tau+\frac{\beta+\alpha x}{\underline{w}}}$, then $\frac{1}{\tau} > \frac{1+h}{\tau+\frac{\beta+\alpha x}{w}}$ for any $w < \underline{w}$. Proposition 8 follows directly. □

Similarly, we can show that

Proposition 9. *If some household with wealth $\overline{w}$ chooses to educate, then all households with $w > \overline{w}$ also choose to do so.*

Note that since in equilibrium each household chooses at least one alternative, we have

Proposition 10. *In an interior equilibrium, there exists an indifferent household $\tilde{w}$ such that all households with $w > \tilde{w}$ educate, while all households with $w < \tilde{w}$ choose not to.*

We can then find the price of education that materializes in an interior equilibrium as

$$p_e = \beta + \alpha F(\tilde{w}), \text{ where } \tilde{w} \text{ is the indifferent household}$$

And, by the definition of an indifferent household, we must have

$$\frac{1}{\tau} = \frac{1+h}{\tau + \frac{\beta+\alpha F(\tilde{w})}{\tilde{w}}}, \text{ or } \tau h\tilde{w} = \beta + \alpha F(\tilde{w}) \tag{2}$$

Proposition 11. *In an interior equilibrium, $\tau h\tilde{w} = \beta + \alpha F(\tilde{w})$*

Note that such an equation can have multiple solutions, which means that there might be multiple equilibria in the economy. In what follows, I will consider two specific cases.

2.6.1 Uniform Distribution

In the case of a uniform distribution of income over $[0, \hat{w}]$ combined with the case of an interior equilibrium, equation 2 simplifies to

$$\tau h \tilde{w} = \beta + \alpha \frac{\tilde{w}}{\hat{w}}, \text{ or } \tilde{w} = \frac{\beta}{\tau h - \frac{\alpha}{\hat{w}}} \tag{3}$$

which is indeed interior if $\tau h \hat{w} \geqslant \alpha + \beta$. The share of educating households is then $\frac{\beta}{\tau h \hat{w} - \alpha}$. Note that the total amount of human capital in the economy then is:

$$HC = \frac{\beta}{\tau h - \frac{\alpha}{\hat{w}}} \cdot \frac{1}{\tau} + \int_{\frac{\beta}{\tau h - \frac{\alpha}{\hat{w}}}}^{\hat{w}} \frac{1 + h}{\tau + \frac{\beta + \alpha \frac{\beta}{\tau h - \frac{\alpha}{\hat{w}}}}{w}} dw \tag{4}$$

This surprisingly actually has a closed-form solution, but that looks very ugly. I guess this means that I did not yet master the skill of creating elegant models, but let us hope that I might do that in the following 8 months. However, for now it is easier to just consider that expression of total human capital graphically.

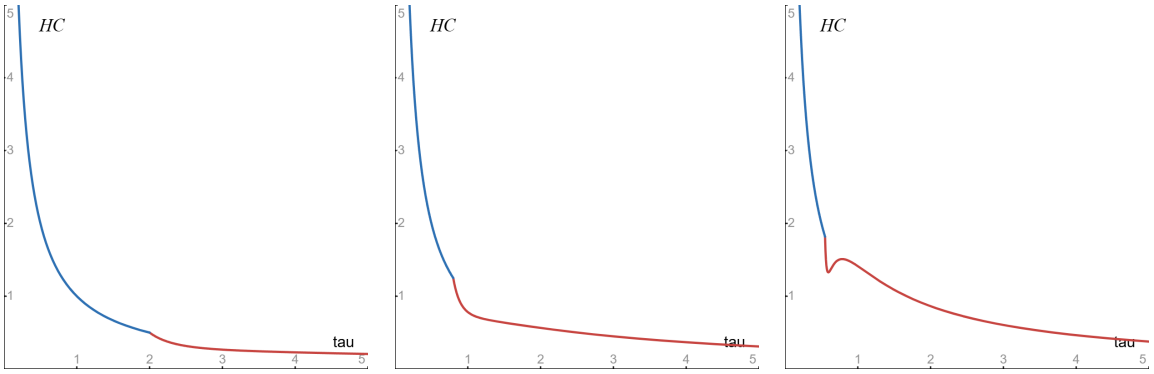


Figure 4: Total Human Capital for the Uniform Case

Figure 4 shows three cases of the total human capital as a function of the cost of children τ. The blue line represents the part where every household chooses not to educate, while the red line represents the part where some share educates. The left graph shows the case of a high α and β. The middle graph shows a case of lower alpha and lower beta. Finally, the right picture shows the case of a high alpha and a lower beta.

The most interesting case for me is the rightmost graph here, as we can see a part of the function where total human capital in the economy *decreases* with a decrease in the cost of child nurture. Similarly to Proposition 6, this result might call for caution.

Proposition 12. *In some cases, it is possible to decrease the total human capital economy by unreasoned attempts to increase it by cheapening the direct cost of child nurture.*

2.6.2 Exponential Distribution

Let us consider Proposition 11 in more detail for the case of an exponential distribution. This might make sense, as literature shows that the actual distribution of wealth might be somewhat similar to an exponential one (Drăgulescu and Yakovenko, 2001).

$$\tau h\tilde{w} = \beta + \alpha(1 - e^{-\lambda\tilde{w}})$$

It would be a mess to consider that analytically in closed form, so let us just note that there will always be at least one equilibrium, as the LHS starts at 0 and goes to infinity, while the RHS starts above zero and has a finite limit. We can also show that the higher τ is, the less people choose to educate. Hence, for the exponential case I cannot make explicit predictions about the total human capital, but I still can state for sure the following:

Proposition 13. *In a world with an exponential distribution of wealth, similar to most distributions, a change in the cost of education does not only affect fertility, but definitely also affects the composition of the population in terms of the share of educated children.*

3 Conclusion

In the model above, I show how the basic quantity-quality tradeoff changes if externalities and norms are included. By modeling the cost of education as dependent on neighbors' choices, I show that the decision to educate children is not made in isolation but is strategically linked within a community.

My key findings are threefold. First, I show that the presence of negative externalities can lead to multiple equilibria, where a society can become stuck in a trap where no one educates their children, even when a better, educating equilibrium is available. Second, I point out potential pitfalls, where well-intentioned policies reducing the direct cost of having children (τ) backfire. Such policies may decrease the total human capital by either shifting the economy from a good to a bad equilibrium or, in a stratified society, by increasing the negative externalities from the poor to the rich so much that it reduces the fertility of the educated class. In any case, such policies always generate changes in the demographic composition of the economy. Finally, I demonstrate that my model is robust and can be generalized to a continuum of agents with heterogeneous incomes.

As for policy implications, my examples of potential failures might call for caution in designing fertility-inducing mechanisms. We often do not understand the full picture with respect to fertility as, for example, we do not understand why the removal of the one-child policy in China did not increase fertility sufficiently. Hence, one might want to advise not to impose laws or try to shift norms in such sensitive matters like household choices regarding fertility.

There are several potential prospects to extend my work from this project. First, it would definitely be helpful to make the model more elegant at the end, as it would

increase the readability significantly, if I added some direct derivations of the total human capital within an economy. Second, it might be interesting to slightly extend the model in a different direction, namely, allow for a continuous education rather than two discrete choices. Finally, and in my opinion most importantly, it would help the model a lot if I were to find any empirical evidence for the effects of neighbors' quality on a household's education and fertility choice. One promising prospect here could be to exploit the Syrian Crisis in 2014 with poor uneducated children randomly entering neighborhoods.

Note on AI Usage

As always, I used various AI for literature suggestions. I used Overleaf-inbuilt GPT for grammar, text revision, structure, and cleaning suggestions. I also asked ChatGPT to write an Abstract as well as a structure for the Conclusion of the project, as I find these are tasks where a clear structure is more important than human creativity. Obviously, I revise and adjust any text or structure suggestions before implementing them into work.

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